

5. CONCLUSIONS

The usual statistical analysis performed on RRAM experimental data in order to characterize the device variability makes use of Weibull distribution. Nevertheless, sometimes the fit obtained to measured data is not accurate. This fact suggests that other statistical distributions could work in a better manner. In this respect, a new methodology is developed in our manuscript by considering phase-type distributions to fit the V_{reset} distribution, the voltage corresponding to the reset processes where RRAM conductive filaments rupture happens. The phase-type distribution class employed can be considered a general class, given that any non-negative distribution can be approximated as needed through a phase-type distribution. From the general phase-type distribution parameter estimation performed on RRAM experimental measurements, the best fit is obtained and it was found that Erlang distribution, a distribution belonging to phase-type distribution class, is particularly appropriate. The phase-type parameters were estimated from experimental data and interpreted from the physically based viewpoint. The first parameter is the number of sequential degradation stages up to the reset and the inverse of the second parameter is the mean V_{reset} in each stage. In addition, the fit is compared with the usual Weibull distribution to shed light on the issue.

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